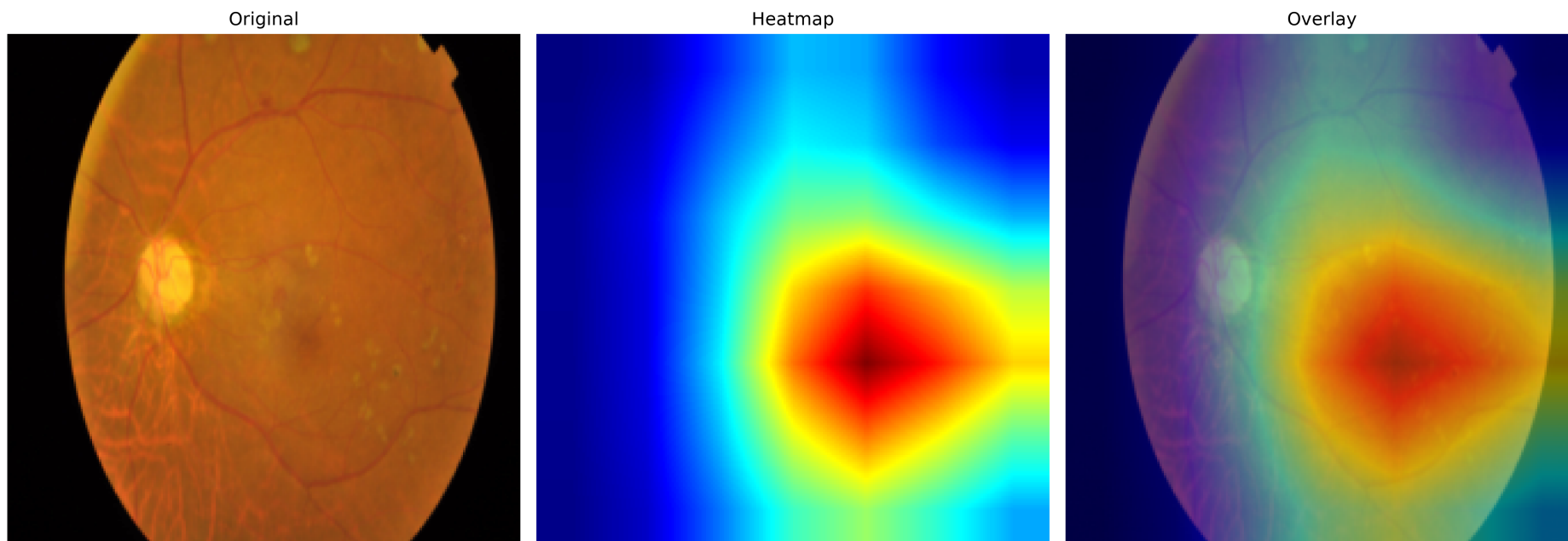


# **FusionMedAI: XAI Failure Analysis**

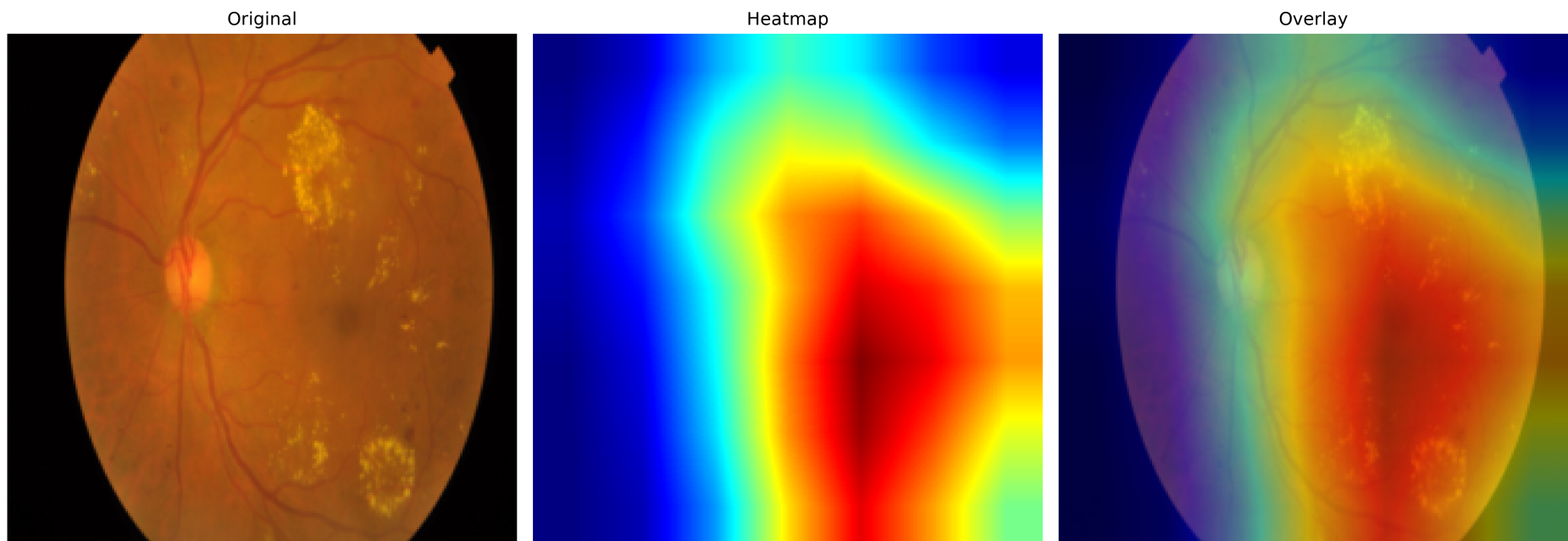
## XAI Visualizations Gallery



Ground Truth: Proliferative Prediction: Moderate Confidence: 0.9793  
Entropy: 0.1074 Reason: Severe confusion example (>1 grade error)

Observation: Attention is highly localized to a single region.

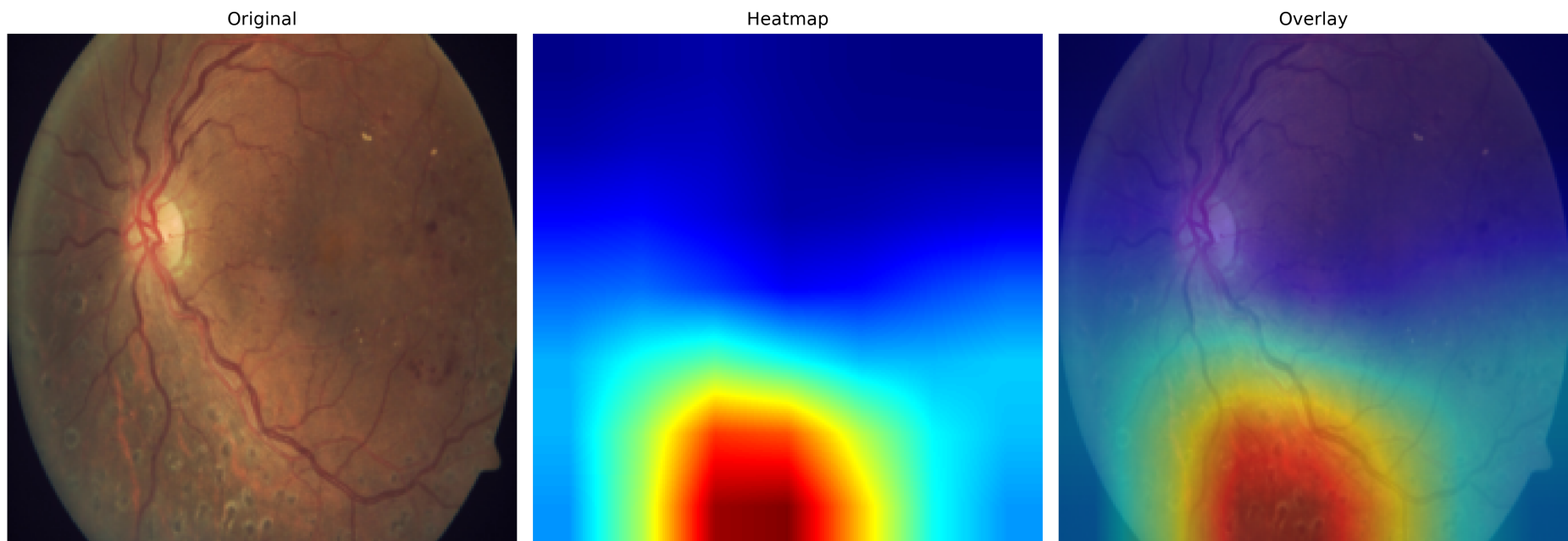
Figure 1: image\_1c4f3aa4df06 XAI Analysis



Ground Truth: Severe    Prediction: Moderate    Confidence: 0.9544  
Entropy: 0.2014    Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention is widely dispersed across retinal tissue.

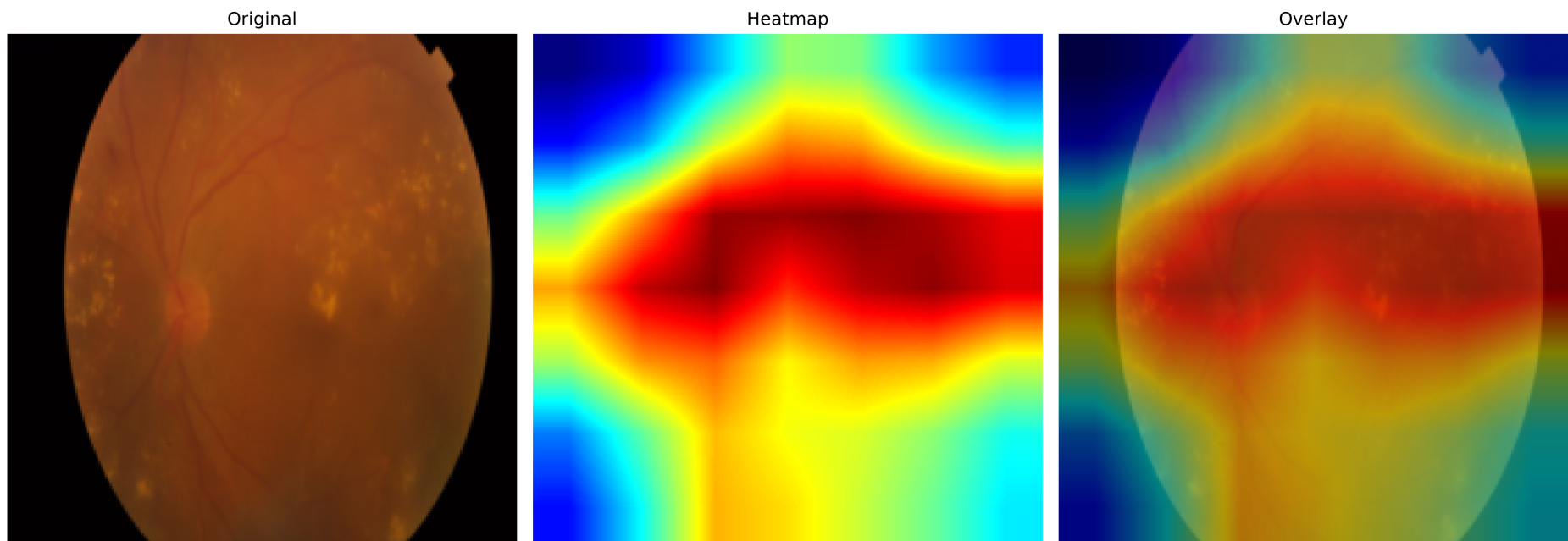
Figure 2: image\_24b87f744598 XAI Analysis



Ground Truth: Severe   Prediction: Proliferative   Confidence: 0.9977  
Entropy: 0.0179   Reason: Most confident incorrect

Observation: Attention is highly localized to a single region.

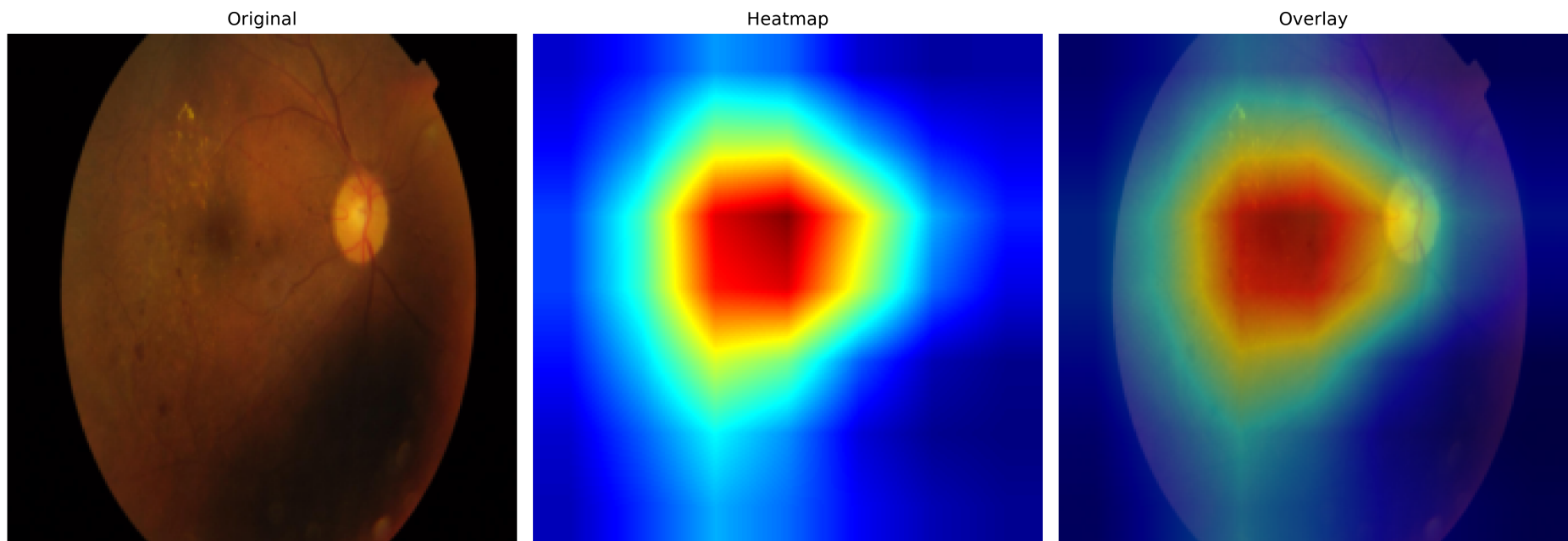
Figure 3: image\_3435fd8675a2 XAI Analysis



Ground Truth: Moderate Prediction: Severe Confidence: 0.9888  
Entropy: 0.0694 Reason: Failure example (Grade Moderate)

Observation: Attention is highly localized to a single region. Attention is widely dispersed across retinal tissue. Attention concentrated near the central (macular) region.

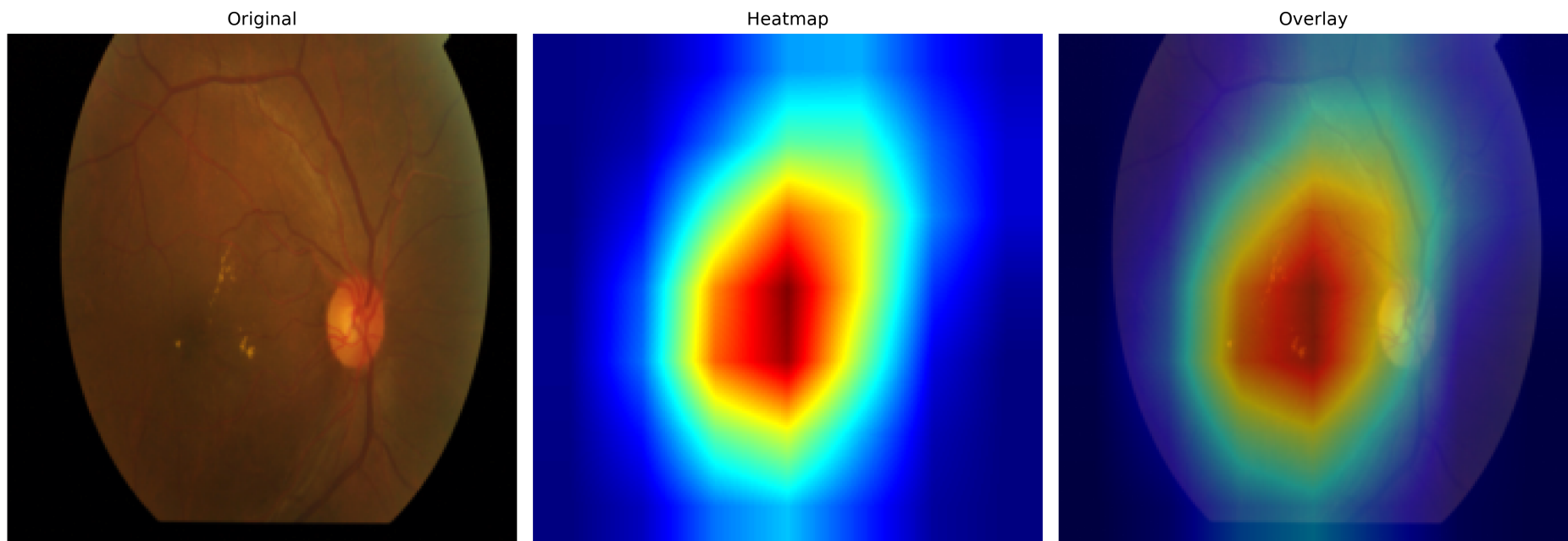
Figure 4: image\_4a44cc840ebe XAI Analysis



Ground Truth: Moderate   Prediction: Severe   Confidence: 0.5346  
Entropy: 1.1635   Reason: Highest entropy prediction

Observation: Attention is highly localized to a single region. Attention concentrated near the central (macular) region.

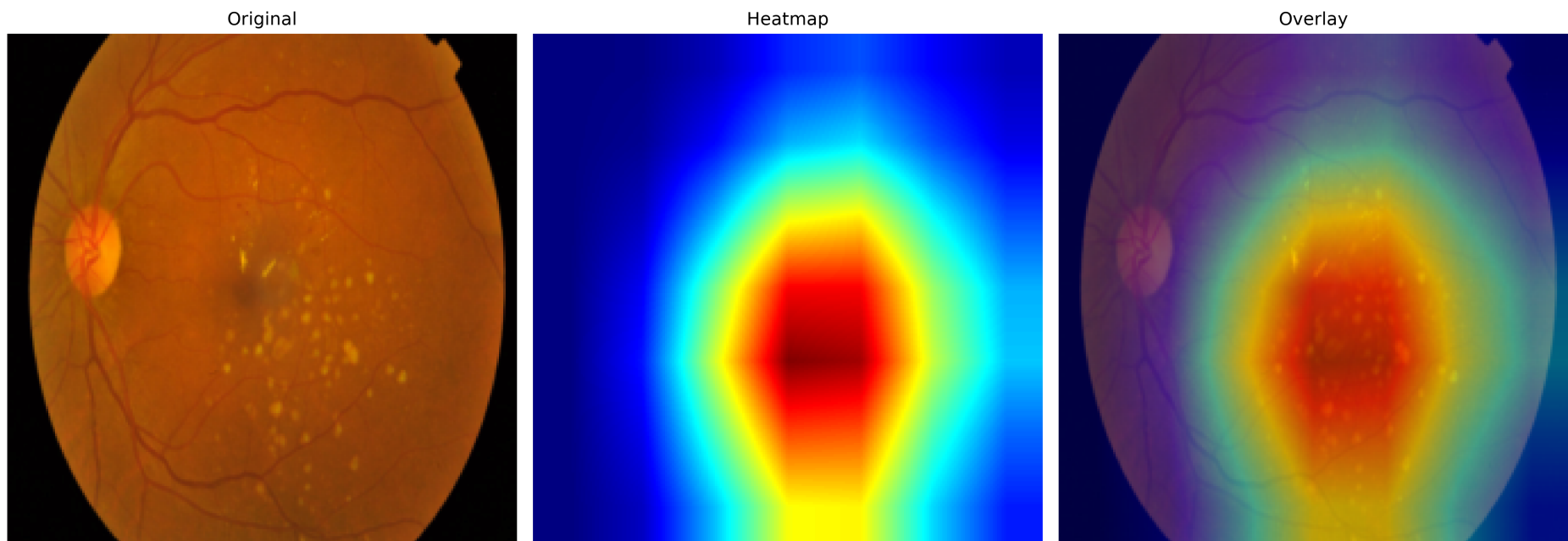
Figure 5: image\_541db13517e2 XAI Analysis



Ground Truth: Mild   Prediction: Moderate   Confidence: 0.7387  
Entropy: 0.7693   Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention concentrated near the central (macular) region.

Figure 6: image\_6c6efb6b1358 XAI Analysis

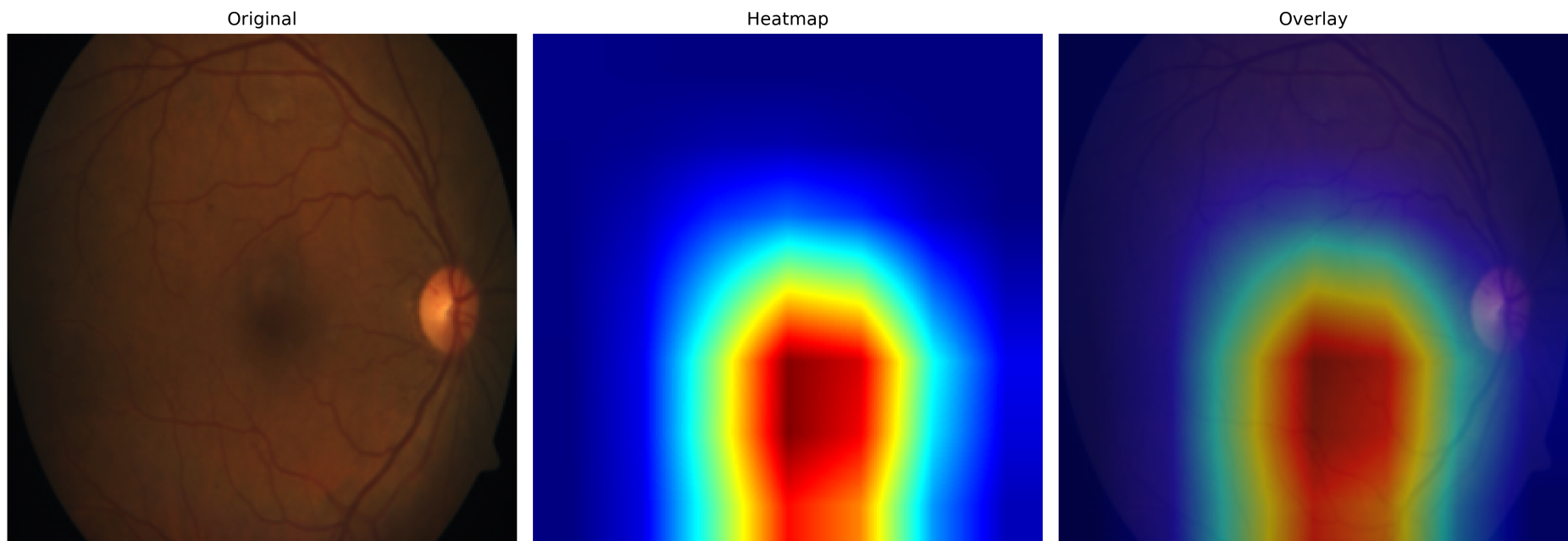


Ground Truth: Mild   Prediction: Moderate   Confidence: 0.9924  
Entropy: 0.0510   Reason: Failure example (Grade Mild)

Observation: Attention is highly localized to a single region. Attention concentrated near the central (macular) region.

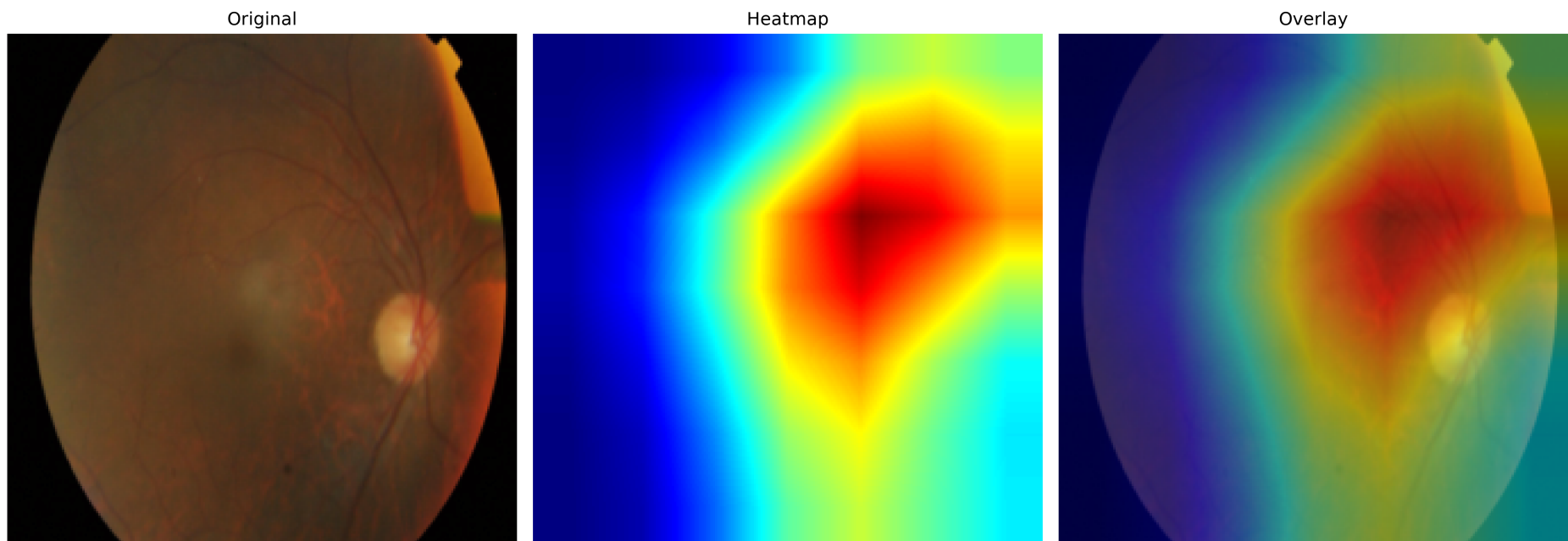
Figure 7: image\_89ed6a0dd53f XAI Analysis





Ground Truth: Mild   Prediction: No DR   Confidence: 0.4686  
Entropy: 1.0865   Reason: Least confident incorrect  
Observation: Attention is highly localized to a single region.

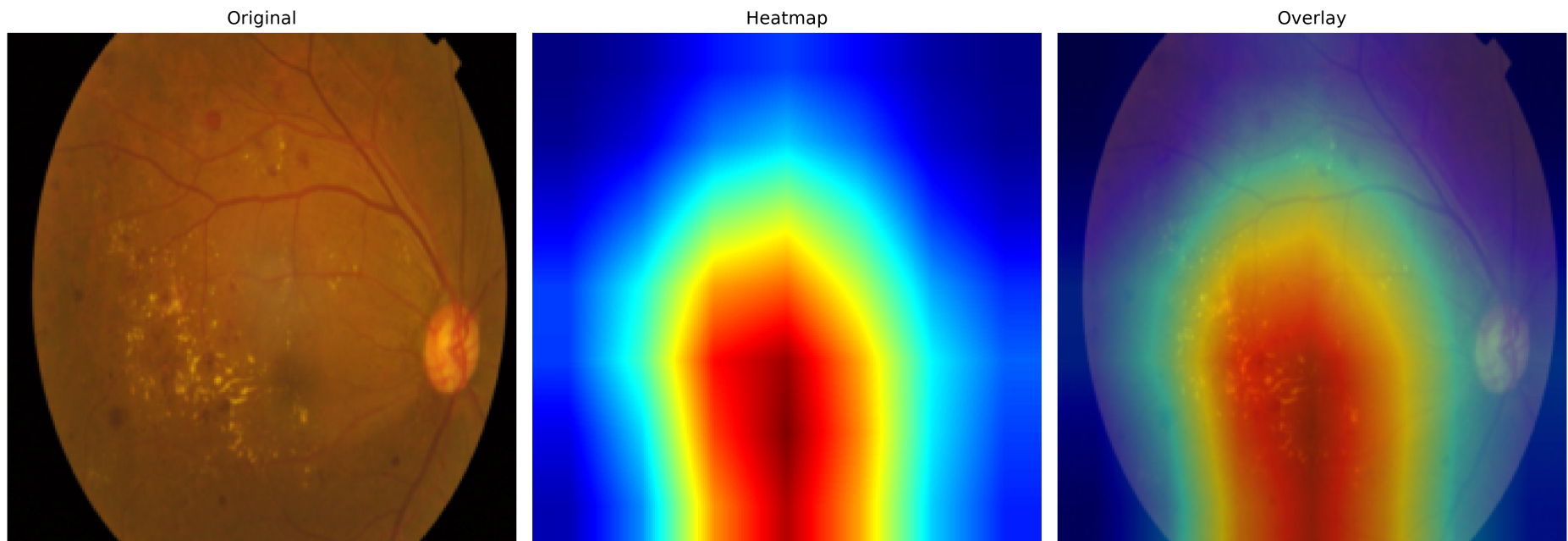
Figure 8: image\_d06ccd0cf4b8 XAI Analysis



Ground Truth: No DR   Prediction: Mild   Confidence: 0.9807  
Entropy: 0.1113   Reason: Failure example (Grade No DR)

Observation: Attention is highly localized to a single region. Attention is widely dispersed across retinal tissue.

Figure 9: image\_da6389d129aa XAI Analysis



Ground Truth: Proliferative Prediction: Severe Confidence: 0.9926  
Entropy: 0.0480 Reason: Failure example (Grade Proliferative)  
Observation: Attention is highly localized to a single region.

Figure 10: image\_f5e6226bd2e0 XAI Analysis